Shubham Raj

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Professional Summary

Data Engineer with 3+ years of experience in enterprise-scale cloud migrations, ETL pipeline development, Dashboarding and EDW modernization on GCP. Proven ability to orchestrate production-grade pipelines, optimize complex SQL transformations, and build Data Vault models.

Experience

Data Engineer | Quantiphi Analytics Solutions Pvt Ltd. *Jan 2024 – Present*

- Led a team of data engineers, structuring workloads, reviewing code, resolving project-level issues, and overseeing parallel development of multiple data models and tables.
- Engineered data pipelines across **healthcare, insurance, finance, and enterprise** datasets.
- Designed and developed **Data Vault models (Raw, Business, Gold layers)** using DBT (Data Build Tool) for transformations, improving analytical scalability and long-term maintainability.
- Built and operated **ETL pipelines** migrating data from **Exasol, Teradata, Mainframe (COBOL)** to **BigQuery**.
- Developed **Apache Airflow (Cloud Composer)** DAGs for orchestration and monitoring across Dev, QA, and Prod environments.
- Optimized complex SQL queries, reducing execution time from **12+ hours to under 10 minutes**.
- Developed **PySpark** transformation jobs triggered by Google Cloud Run and executed via Dataproc Serverless.
- Implemented DAG-level **alerting, monitoring, and failure reporting** to improve SLA compliance.
- Led **data validation and reconciliation** for large-scale migrations, ensuring record-level accuracy and zero data loss.
- Converted legacy **COBOL programs, LUA and Teradata** scripts into optimized BigQuery SQL.
- Analyzed large datasets using BigQuery, building **Tableau** dashboards and visual reports that supported business **decision-making and operational monitoring**.
- Applied **clean architecture** and security practices using parameterization, YAML configs, secrets management, and CI/CD.

Data Engineer - Internship | Quantiphi Analytics Solutions Pvt Ltd. *Jan 2023 – Jan 2024*

- Built and deployed end-to-end data pipelines for **Snowflake to BigQuery** migration project, using **Informatica IICS**.
- Developed Tableau dashboards and analytical reports for business stakeholders.
- Performed SQL analysis, query optimization, and validated data across Dev and QA.

Web Developer - Internship | Dr. Reddy's Laboratories Pvt Ltd. *Jul 2021 – Nov 2021*

- Built APIs to fetch and display millions of records from BigQuery for analytical dashboards, reducing cloud compute costs.
- Worked on user authentication, authorization, and state management. Implemented caching and optimized BqSQL queries.

Projects

Claims Automation Platform (Freelance)

- Developed scalable backend APIs using **ASP.NET Core 8.0** following **SOLID and MVC principles**.
- Designed database schemas and optimized stored procedures in **MS SQL Server**.
- Automated claims workflows, reducing average turnaround time by **90%**.

Technical Skills

Data Engineering: ETL/ELT, Data Pipelines, Data Migration, Data Modeling, Data Vault, EDW Modernization, Dashboarding
Cloud & Big Data: Google Cloud Platform, BigQuery, Cloud Storage, Dataproc, Cloud Run, PySpark / Apache Spark
Programming: Python, SQL, COBOL (Legacy), LUA
Orchestration & DevOps: Apache Airflow, Cloud Composer, DBT, CI/CD, Git, GitHub, GitHub Actions, Bitbucket
Databases: Exasol, Teradata, Mainframe, Snowflake, BigQuery, MySQL, MS SQL Server

Certifications

Generative AI for Developers - Multiple Badges | Google Cloud Badge *Feb 2024 – Sep 2024*
Professional Data Engineer | Google Cloud Certified *Dec 2024*
Oracle Cloud - Foundations Associate | Oracle Certified *Nov 2023*
Oracle Cloud - AI Foundations Associate | Oracle Certified *Nov 2023*
Associate Cloud Engineer | Google Cloud Certified *May 2023*

Education

National Institute of Technology, Durgapur *Jul 2019 – Jun 2023*

- B.Tech - Metallurgical and Materials Engineering